

WHAT I DID

2-Week Software Engineering Shadow

Shadowed a senior software engineer for two weeks in a hybrid format - in-person office days and remote distributed team simulation via Slack and Zoom. Learned the full software development lifecycle from ideation to deployment.

The Timeline

- Day 1 - Hello World**
Dev environment setup. Created first GitHub repo. Built and deployed a website.
- Day 2 - Real Engineering**
Observed standup and sprint planning. Whiteboarded web architecture and database design.
- Day 3 - Ideation**
Brainstormed app ideas with AI. Evaluated 3 finalists. Selected Bramble.
- Weekend - JavaScript**
freeCodeCamp JavaScript course. Variables, loops, conditionals, functions.
- Remote Day - Distributed Work**
Full remote simulation. Standups on Zoom. Slack threading. End-of-day reports.
- Architecture Day**
Wireframes in Excalidraw. Tech stack decisions. Database schema with 5 tables.
- Build Days**
Linear project management. Auth, database, Vercel deployment. 5 PRs merged.
- Final Sprint**
AI plant ID, community map, journal, invasive species detection. Shipped to bramblemap.com.

Tools I Used

Cursor IDE

Claude AI

GitHub

Linear

Slack

Zoom

TypeScript

React

Next.js

Tailwind CSS

Supabase

PostGIS

Drizzle ORM

Vercel

Excalidraw

freeCodeCamp

Skills Developed

JavaScript Fundamentals

Git & GitHub Workflows

AI-Assisted Engineering

Project Management (Linear)

Database Schema Design

Auth & Deployment

Distributed Team Work

Professional Communication

Practiced

Practiced

Internalized

Practiced

Exposed

Practiced

Practiced

Internalized

Internalized

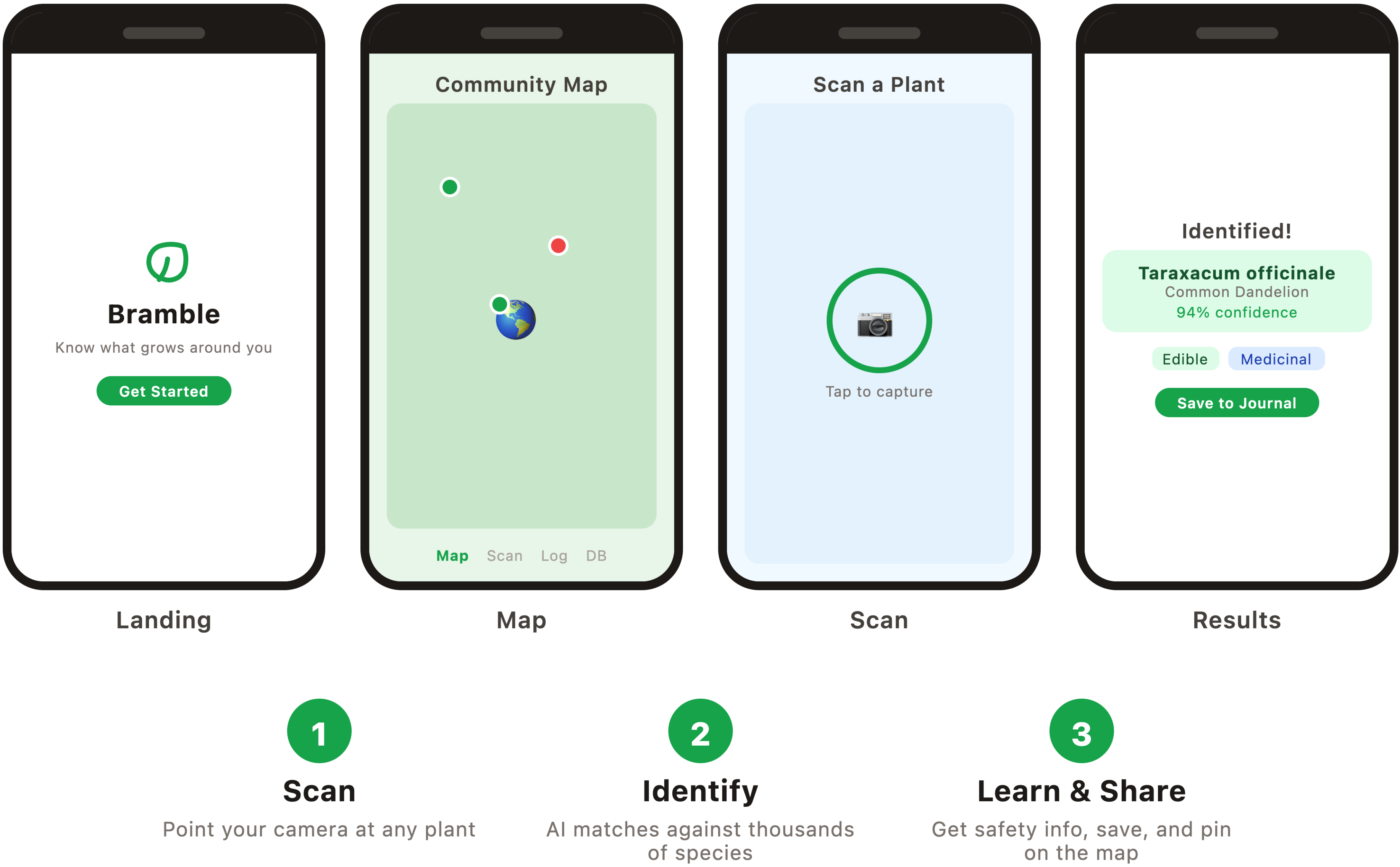
Practiced

Exposed

Bramble

COMMUNITY FORAGING APP

Identify wild plants with AI, keep a foraging journal, and share discoveries on a community map. Built for foragers, hikers, and the naturally curious.



Features

- AI Plant Identification**
Snap a photo and Bramble uses Pl@ntNet to identify the species with confidence scoring.
- Safety Information**
Every plant shows edible, medicinal, toxic, and invasive status. Unverified plants show a clear warning.
- Invasive Species Detection**
Cross-references 1,500+ species from the USGS national invasive species registry.
- Community Map**
Interactive map shows plant discoveries from all users within 25 miles using PostGIS spatial queries.
- Foraging Journal**
Every scan auto-creates an entry with species, GPS, date, and confidence. Add personal notes.
- Plant Database**
Search 34,000+ species from the USDA database. Filter by edible, medicinal, toxic, or invasive.
- Secure Authentication**
Google OAuth via Supabase. No passwords to remember.
- 3 Data Sources**
Pl@ntNet API (real-time ID), USDA PLANTS (34k species), USGS US-RIIS (invasive registry).

Tech Stack

Next.js 16 / React 19 / TypeScript / Tailwind CSS / Supabase + PostGIS / Drizzle ORM / Vercel

Noah Madsen
Senior Capstone, 2026

WHAT I LEARNED

Reflections from a 2-week engineering shadow

- "Asking for help is engineering, not weakness."**
The 20-minute rule: try seriously for 20 minutes, then escalate. Going dark when stuck makes the problem compound while everyone assumes work is happening.
- "AI proposes, human curates."**
AI-assisted coding is a multiplier on what you already understand, not a substitute for understanding. The human judgment layer is where engineers earn their keep.
- "Infrastructure is invisible to users."**
Building the foundation feels productive. But real engineering is shipping end-to-end. Deciding what to cut matters as much as deciding what to build.
- "Show me where in MY code this is referenced."**
When an AI asks for something you don't recognize, ask it to prove the requirement by reading the actual code.
- "Match your tools to the shape of the work."**
Cycles for ongoing teams. Milestones for projects with an end date. Default settings aren't always right.
- "The next engineer should run it without asking a single question."**
If a machine-specific thing is required, that requirement needs to be documented. That's the difference between code and a product.

The Biggest Takeaway

"Senior engineers don't ship perfect code. They ship code, then iterate. The skill I needed most wasn't coding - it was saying 'I'm stuck, here's what I tried, can you help?'"

Try It Live



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